SQL Code

```sql
use ecommerce; -- use database ecommerece
select * from ecommerce_orders;-- retrieve all the recordss from the table ecommerce table under ecommerce database to know the column names and in prder to perform required tasks
/* select total revenue generated*/
select sum(price*quantity) as total_revenue
from ecommerce_orders;
-- Show Total Orders by Payment Method

select sum(order_id),payment_method
from ecommerce_orders
group by payment_method;
-- 3. Find Revenue by Product Category
SELECT payment_method, COUNT(*) AS total_orders
FROM ecommerce_orders
GROUP BY payment_method;
-- 4. Find Top 5 Customers by Spending
select sum(price*quantity) as total_revenue
from ecommerce_orders
group by product_category;
-- 4. Find Top 5 Customers by Spending
select  customer_name, sum(price*quantity)as total_spent
from ecommerce_orders
group by customer_id,customer_name
order by total_spent desc
limit 5;
-- Count Orders by Order Status
select order_status,count(*)as total_orders
 from ecommerce_orders
 group by order_status;
 -- Find Average Delivery Days
select avg(delivery_days) as avg_delivery_days
 from ecommerce_orders ;
-- Find Most Sold Product
select product_name,sum(quantity) as count_of_product
from ecommerce_orders
group by product_name
order by product_name
 limit 3;
 -- Find Average Rating by Product Category
select product_category,avg(rating) as avg_of_pro_cat
from ecommerce_orders
 group by product_category;
 -- Find Total Orders in Each City
select city,sum(quantity) as total_product from ecommerce_orders group by city;
-- Find Revenue Generated from Each City
select city,sum(price*quantity) as total_revenue from ecommerce_orders group by city ;
-- Find Customers Who Used COD Payment
SELECT customer_name, product_name, payment_method
FROM ecommerce_orders
```

```sql
WHERE payment_method = 'COD';
--  Find Orders with Rating Greater Than 4.5
SELECT customer_name, product_name, rating
FROM ecommerce_orders
WHERE rating > 4.5;
-- Find Total Orders by Gender
SELECT gender, COUNT(*) AS total_orders
FROM ecommerce_orders
GROUP BY gender;
-- Find Highest Priced Product
SELECT product_name, price
FROM ecommerce_orders
ORDER BY price DESC
LIMIT 1;
-- Find Cancelled Orders
SELECT order_id, customer_name, product_name
FROM ecommerce_orders
WHERE order_status = 'Cancelled';
-- Find Customers from Mumbai
SELECT customer_name, product_name
FROM ecommerce_orders
WHERE city = 'Mumbai';
-- Find Total Quantity Sold by Category
SELECT product_category,
SUM(quantity) AS total_quantity
FROM ecommerce_orders
GROUP BY product_category;
-- Find Orders with Delivery More Than 5 Days
SELECT order_id, customer_name, delivery_days
FROM ecommerce_orders
WHERE delivery_days > 5;
-- Find Average Age of Customers
SELECT AVG(age) AS avg_age
FROM ecommerce_orders;
-- Find Revenue Generated by Each Payment Method
SELECT payment_method,
SUM(price * quantity) AS revenue
FROM ecommerce_orders
GROUP BY payment_method;
```